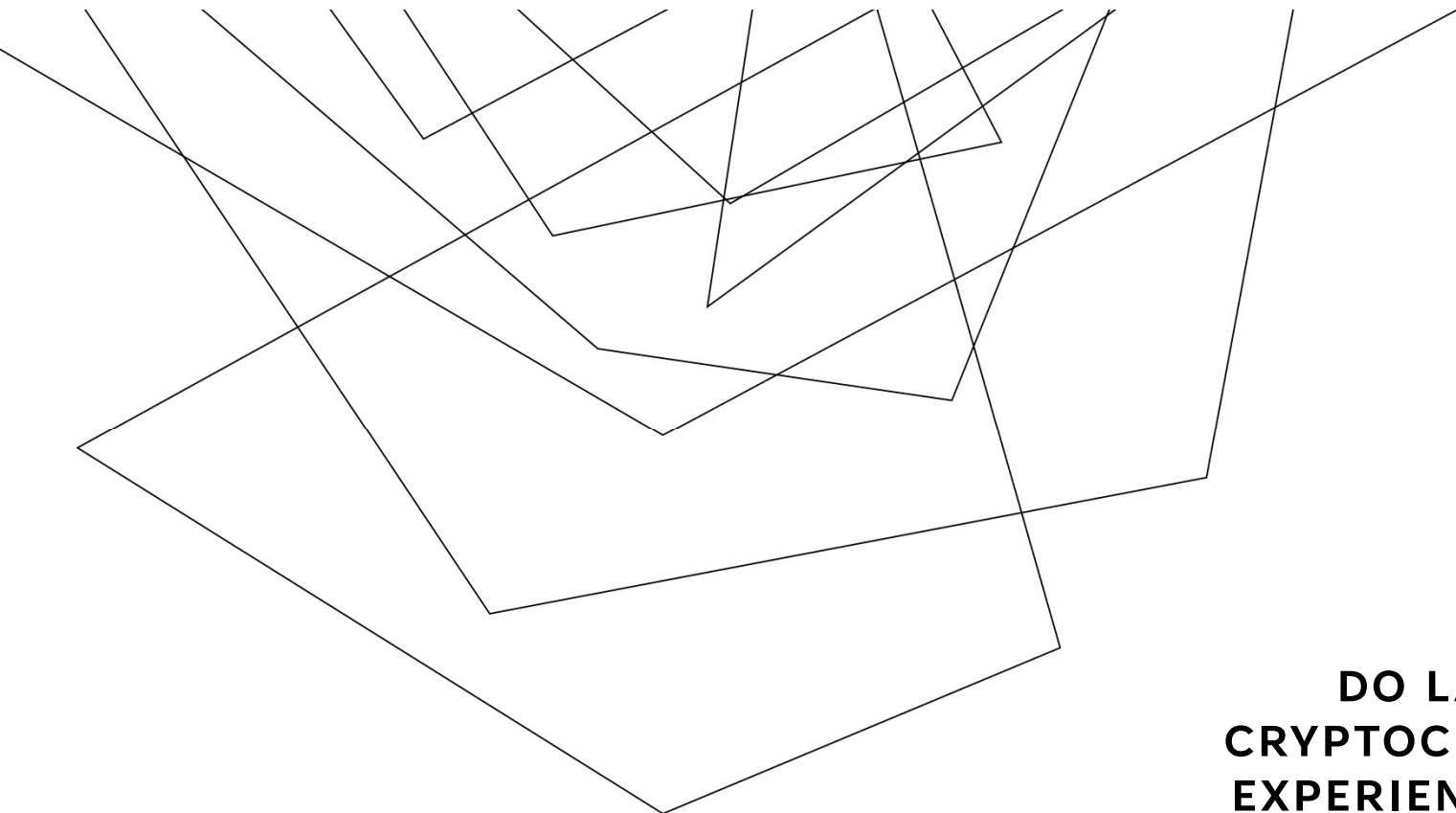




DO LARGER CRYPTOCURRENCIES EXPERIENCE LOWER PRICE VOLATILITY?

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ECO 590 - SPRING 2026

RESEARCH

Research Question: Do cryptocurrencies with larger market capitalizations tend to have lower price volatility?

Motivation: In today's financial system, cryptocurrency markets play a crucial role, and their prices are changing frequently. Market capitalization shows the size of the cryptocurrency, while price volatility shows how much the price has changed. This project will research how these two metrics are related to better understand the differences in risk and stability across cryptocurrencies.

Variables Collected:

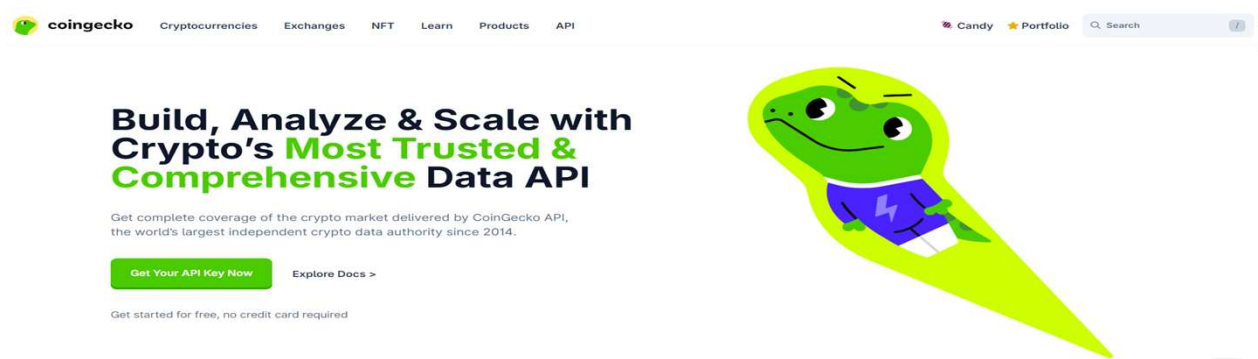
- Cryptocurrency Name
- Current Price
- Market Capitalization
- Trading Volume
- 24-Hour Price Change
- Price Volatility
- Volume to Market Cap Ratio

DATA

Source: CoinGecko API

API Endpoint: I used the CoinGecko `/coins/markets` endpoint, which gave me the current market data for different cryptocurrencies

Difficulty: The hardest part was looking for the correct endpoint because there were various endpoints for different types of data on cryptocurrencies, but once I found the `/coins/markets` endpoint, the data was well organized



DATA COLLECTION (PART 1)

CoinGecko → Coins & Tokens → /coins/markets

The image shows a grid of nine API endpoint cards from CoinGecko. The first card, 'Coins & Tokens', is highlighted with a red border. It includes an icon of a wallet, the endpoint path `/coins/*`, and a description: 'Get real-time and historical prices, market data and metadata for coins and tokens listed on CoinGecko, including images, descriptions, links, social stats and supply info.' The other cards are: 'Global' (world icon, `/global/*`), 'Exchange & Derivatives' (document icon, `/exchanges/*` and `/derivatives/*`), 'On-chain DEX' (DEX icon, `/onchain/*`), 'On-chain Trades' (line chart icon, `/onchain/.../trades/*`), 'Historical Charts' (clock icon, `/coins/.../market_chart/*`), 'OHLC & OHLCV Charts' (OHLCV icon, `/ohlcv/*` and `/ohlcv/*`), 'Search & Discovery' (magnifying glass icon, `/search/*`), and 'Coin Categories' (category icon, `/categories/*`).

Endpoint	Description
Coins & Tokens <code>/coins/*</code>	Get real-time and historical prices, market data and metadata for coins and tokens listed on CoinGecko, including images, descriptions, links, social stats and supply info.
Global <code>/global/*</code>	Get global cryptocurrency market data such as total market cap, volume and dominance over time.
Exchange & Derivatives <code>/exchanges/*</code> <code>/derivatives/*</code>	Get data for centralized spot and derivatives exchanges on CoinGecko, including tickers (trading pairs), volumes, market data and exchange metadata.
On-chain DEX <code>/onchain/*</code>	Get real-time and historical data for onchain pools and tokens listed on GeckoTerminal, including prices, market cap, volume and liquidity.
On-chain Trades <code>/onchain/.../trades/*</code>	Get or stream trades from onchain pools and tokens, including transaction hash, wallet addresses, trade direction, trade size and value.
Historical Charts <code>/coins/.../market_chart/*</code>	Get up to 12 years of historical crypto market data and price charts, with granularity down to every 5 minutes.
OHLC & OHLCV Charts <code>/ohlcv/*</code> <code>/ohlcv/*</code>	Get real-time and historical OHLCV (open, high, low, close, volume) data for building candlestick charts, with up to sub-second granularity.
Search & Discovery <code>/search/*</code>	Get market discovery data such as trending coins, top gainers and losers, new pools and tokens, and search coins, categories and exchanges by name, symbol or contract address.
Coin Categories <code>/categories/*</code>	Get the full list of coin and token categories across CoinGecko and GeckoTerminal, along with category-level market data.

DATA COLLECTION (PART 2)

Looked through the variables & selected the needed ones

```
[
  {
    "id": "bitcoin",
    "symbol": "btc",
    "name": "Bitcoin",
    "image": "https://coin-images.coingecko.com/coins/images/1/large/bitcoin.png?1696501400",
    "current_price": 80208,
    "market_cap": 1605847553832,
    "market_cap_rank": 1,
    "fully_diluted_valuation": 1605847553832,
    "total_volume": 34455920182,
    "high_24h": 82041,
    "low_24h": 79880,
    "price_change_24h": -1345.58070331591,
    "price_change_percentage_24h": -1.64994,
    "market_cap_change_24h": -26406621864.9063,
    "market_cap_change_percentage_24h": -1.6178,
    "circulating_supply": 20028418,
    "total_supply": 20028418,
    "max_supply": 21000000,
    "ath": 126080,
    "ath_change_percentage": -36.38355,
    "ath_date": "2025-10-06T18:57:42.558Z",
    "atl": 67.81,
    "atl_change_percentage": 118184.52513,
    "atl_date": "2013-07-06T00:00:00.000Z",
    "roi": null,
    "last_updated": "2026-05-12T17:16:24.549Z"
  },
]
```

DATA COLLECTION (PART 3)

Used Python to request data from API

1. Creating the dataset from the API

```
url = "https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd" #Link of CoinGecko API (where I get the crypto data)

#settings for the API request
params = {
    "vs_currency": "usd", #get prices in US dollars
    "order": "market_cap_desc", #sort coins by biggest market cap first
    "per_page": 50, #gets the top 50 cryptocurrencies, only want to collect top 50 cryptos
    "page": 1, #uses the first page, so the data only includes top 50 cryptocurrencies by market cap
    "price_change_percentage": "24h" #include 24-hour price change %
}
response = requests.get(url, params=params) #sends request to the API and gets the data
data = response.json() #converts the response into JSON format
df = pd.DataFrame(data) #turns the data into a pandas dataframe
```

DATA COLLECTION (PART 4)

5. Making sure the data set was collected correctly

```
df.info() #checks for column names and data types
df.isnull().sum() #checks if any of the columns have missing values
df.describe() #shows the summary statistics to check if the numbers in the data set make sense
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  50 non-null     object
1   current_price                         50 non-null     float64
2   market_cap                            50 non-null     int64
3   total_volume                         50 non-null     float64
4   price_change_percentage_24h          49 non-null     float64
5   price_volatility                     49 non-null     float64
6   volume_to_market_cap_ratio           50 non-null     float64
dtypes: float64(5), int64(1), object(1)
memory usage: 2.9+ KB
```

	current_price	market_cap	total_volume	price_change_percentage_24h	price_volatility	volume_to_market_cap_ratio
count	50.000000	5.000000e+01	5.000000e+01	49.000000	49.000000	50.000000
mean	1903.005926	5.297475e+10	2.726548e+09	-1.588229	1.827960	0.058940
std	11408.641908	2.307392e+11	9.593431e+09	2.101078	1.891692	0.107837
min	0.000006	1.781542e+09	0.000000e+00	-8.632440	0.000000	0.000000
25%	0.751852	2.838563e+09	4.117712e+07	-3.019930	0.149080	0.010139
50%	1.125000	4.406728e+09	1.785854e+08	-1.373480	1.579420	0.024578
75%	53.432500	9.944495e+09	4.582973e+08	-0.003190	3.026580	0.067090
max	80672.000000	1.615346e+12	5.796089e+10	4.808580	8.632440	0.695534

NEW VARIABLES

Price Volatility:

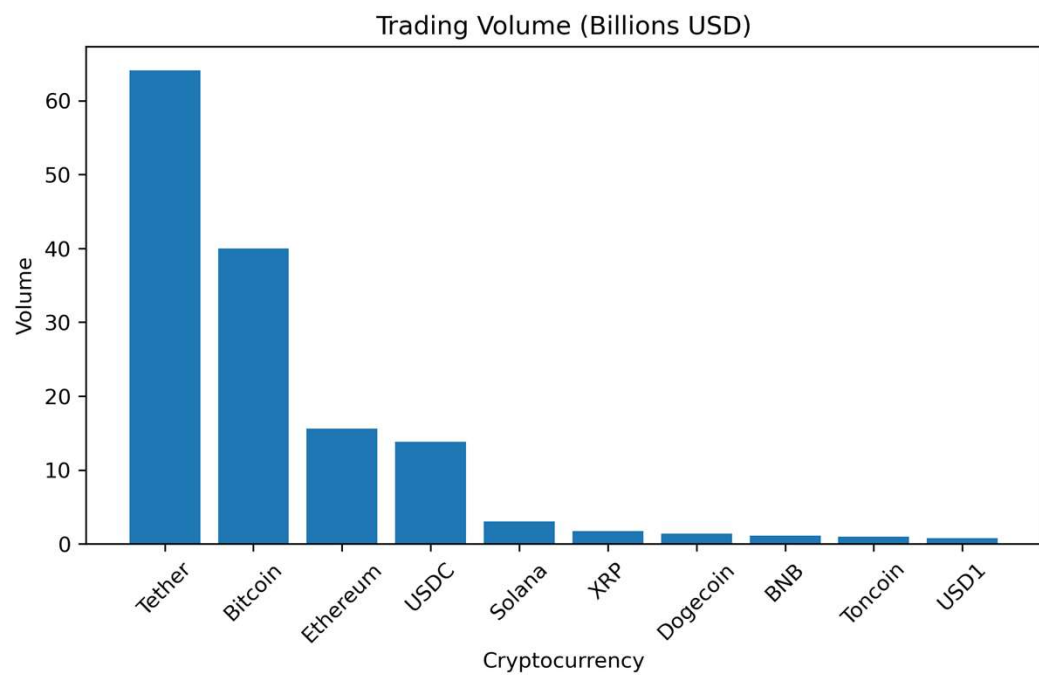
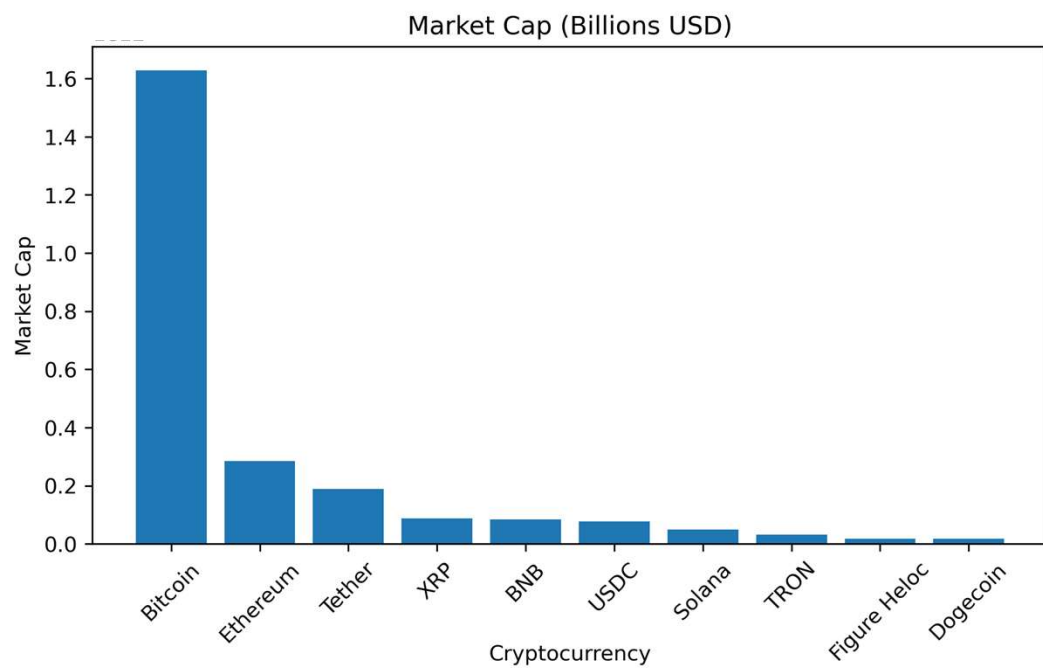
- Measures the absolute value of the 24-hour percent price change
- Helped me focus on how large the price movement was, not whether the price went up or down

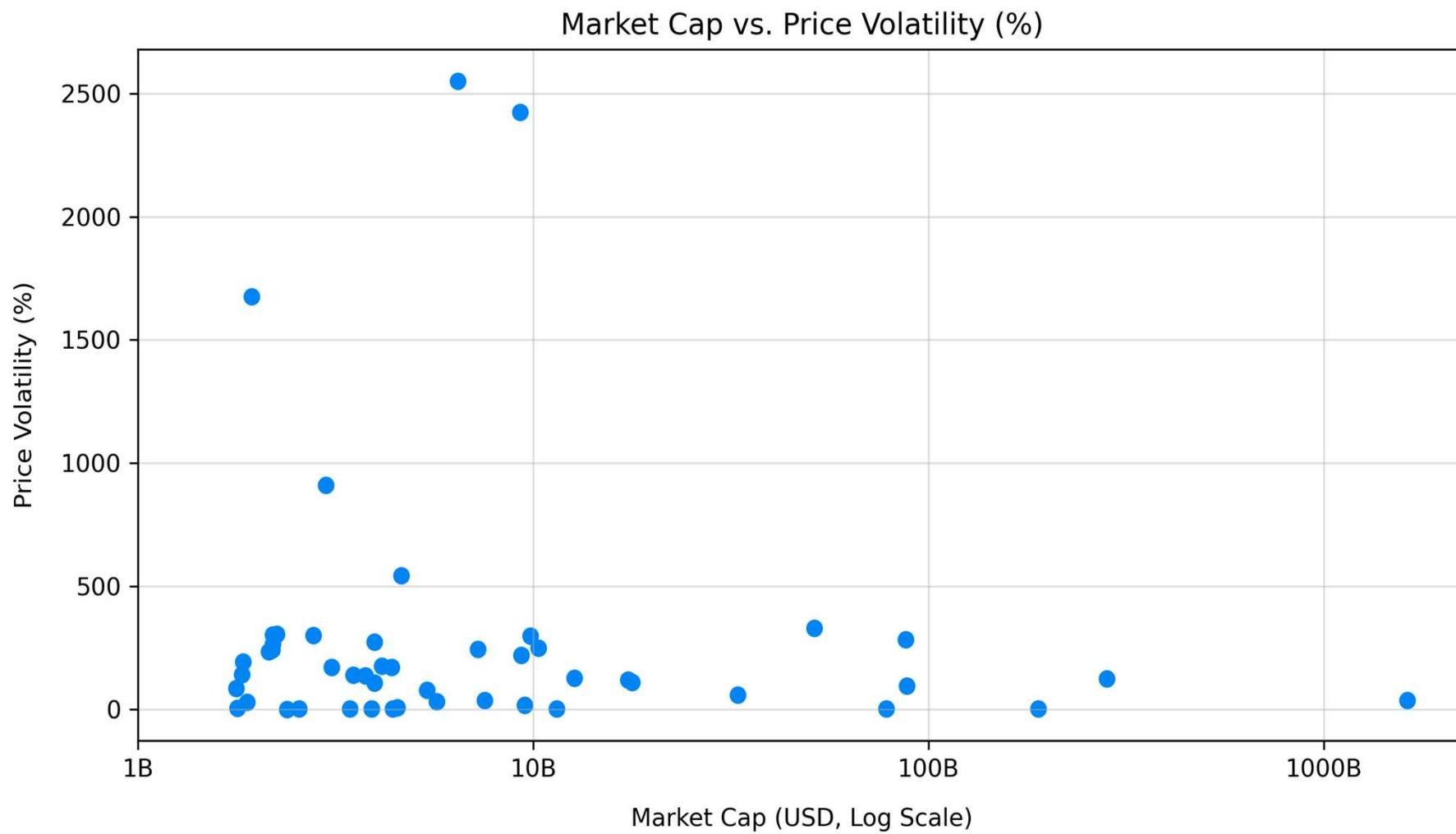
Volume to Market Cap Ratio:

- Divides the trading volume by the market capitalization
- Shows how much trading activity a cryptocurrency has when compared to its overall size
- A higher ratio means that the cryptocurrency is being traded more heavily when compared to its market value

	name	current_price	market_cap	total_volume	price_change_percentage_24h	price_volatility	volume_to_market_cap_ratio
0	Bitcoin	80386.000000	1609870659581	3.459339e+10	-1.46330	1.46330	0.021488
1	Ethereum	2273.850000	274412142391	1.474437e+10	-2.33447	2.33447	0.053731
2	Tether	0.999652	189762732842	5.892913e+10	0.00756	0.00756	0.310541
3	XRP	1.430000	88454428629	2.260380e+09	-2.74507	2.74507	0.025554
4	BNB	654.510000	88221854297	1.129780e+09	-0.78602	0.78602	0.012806
5	USDC	0.999836	77166265384	1.270862e+10	0.01314	0.01314	0.164691
6	Solana	94.400000	54553316828	3.595021e+09	-2.85220	2.85220	0.065899
7	TRON	0.347794	32970233462	4.913829e+08	-0.88628	0.88628	0.014904
8	Figure Heloc	1.032000	18213538966	2.164429e+07	-0.72296	0.72296	0.001188
9	Dogecoin	0.108536	16737659692	1.177740e+09	-2.29127	2.29127	0.070365

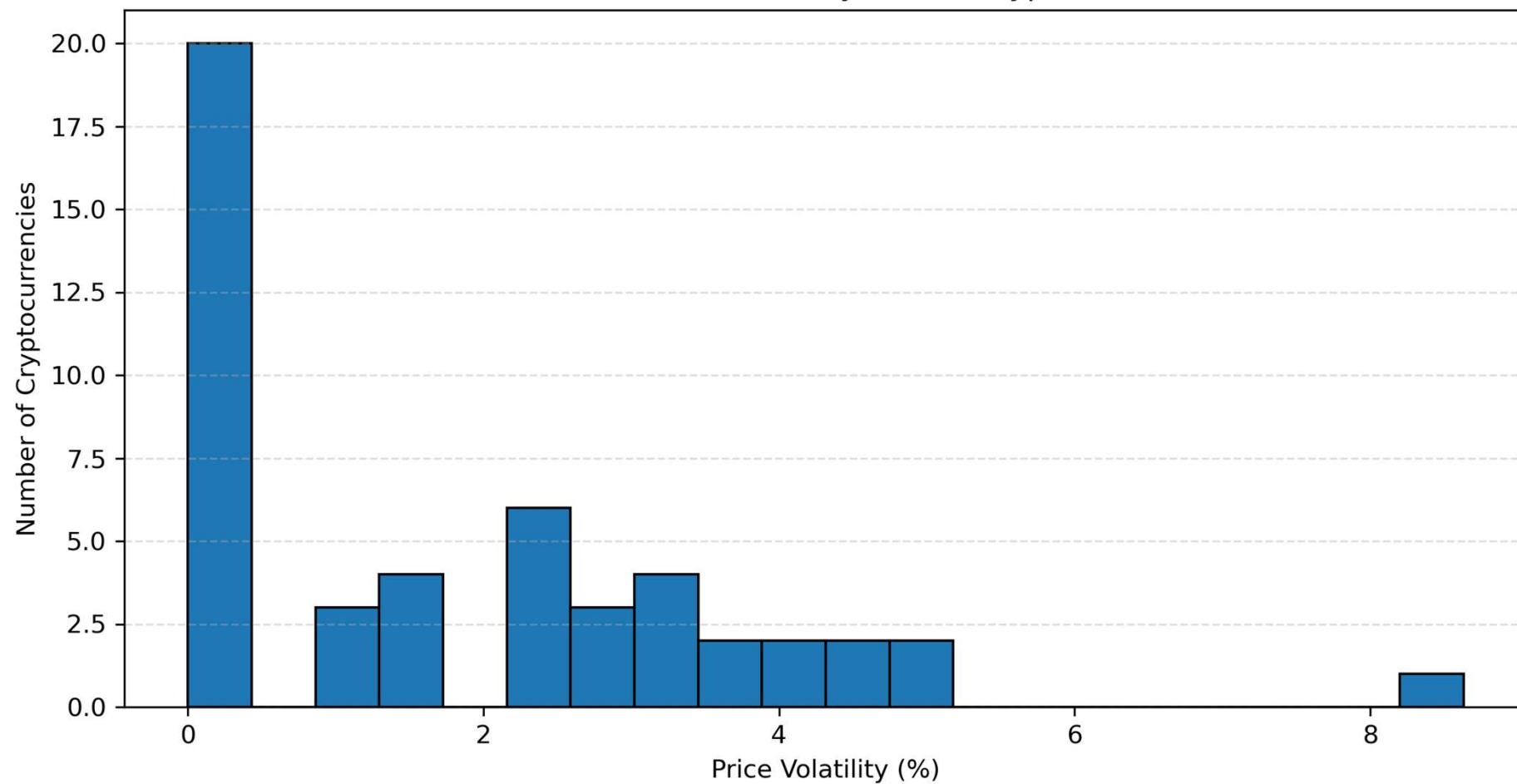
Top 10 Cryptocurrencies: Market Cap vs Trading Volume (Billions USD)





Source: CoinGecko

Distribution of Price Volatility Across Cryptocurrencies



Source: CoinGecko

MODELS

Model 1 (Research Model):

Price Volatility_i = $\beta_0 + \beta_1 \text{Log Trading Volume}_i + \beta_2 \text{Log Market Cap}_i + u_i$

Model 2 (Filtered Model):

Price Volatility_i = $\beta_0 + \beta_1 \text{Log Trading Volume}_i + \beta_2 \text{Log Market Cap}_i + u_i$

Model 3 (Preferred Model):

Price Volatility_i = $\beta_0 + \beta_1 \text{Log Market Cap}_i + \beta_2 \text{Log Trading Volume}_i + \beta_3 \text{Volume-to-Market Cap Ratio}_i + u_i$

Cryptocurrency Regression Results

	Research Model	Price Volatility	
		Filtered Model	Preferred Model
	(1)	(2)	(3)
Log Trading Volume	0.365** (0.164)	0.095 (0.279)	0.295 (0.210)
Log Market Cap	-0.596** (0.257)	-0.121 (0.452)	-0.525* (0.291)
Volume-to-Market-Cap Ratio			1.665 (3.102)
Constant	8.503** (4.201)	2.185 (6.998)	8.101* (4.300)
Observations	48	13	48
R ²	0.118	0.012	0.124
Adjusted R ²	0.079	-0.186	0.064
Residual Std. Error	1.817 (df = 45)	1.262 (df = 10)	1.831 (df = 44)
F Statistic	3.008* (df = 2; 45)	0.059 (df = 2; 10)	2.070 (df = 3; 44)

Note:

* p<0.1; ** p<0.05; *** p<0.01

REGRESSION RESULTS

Research Model: $R^2 = 0.118$

- Log Trading Volume = 0.365, significant at 5%
- Log Market Cap = -0.596, significant at 5%

Filtered Model: $R^2 = 0.012$

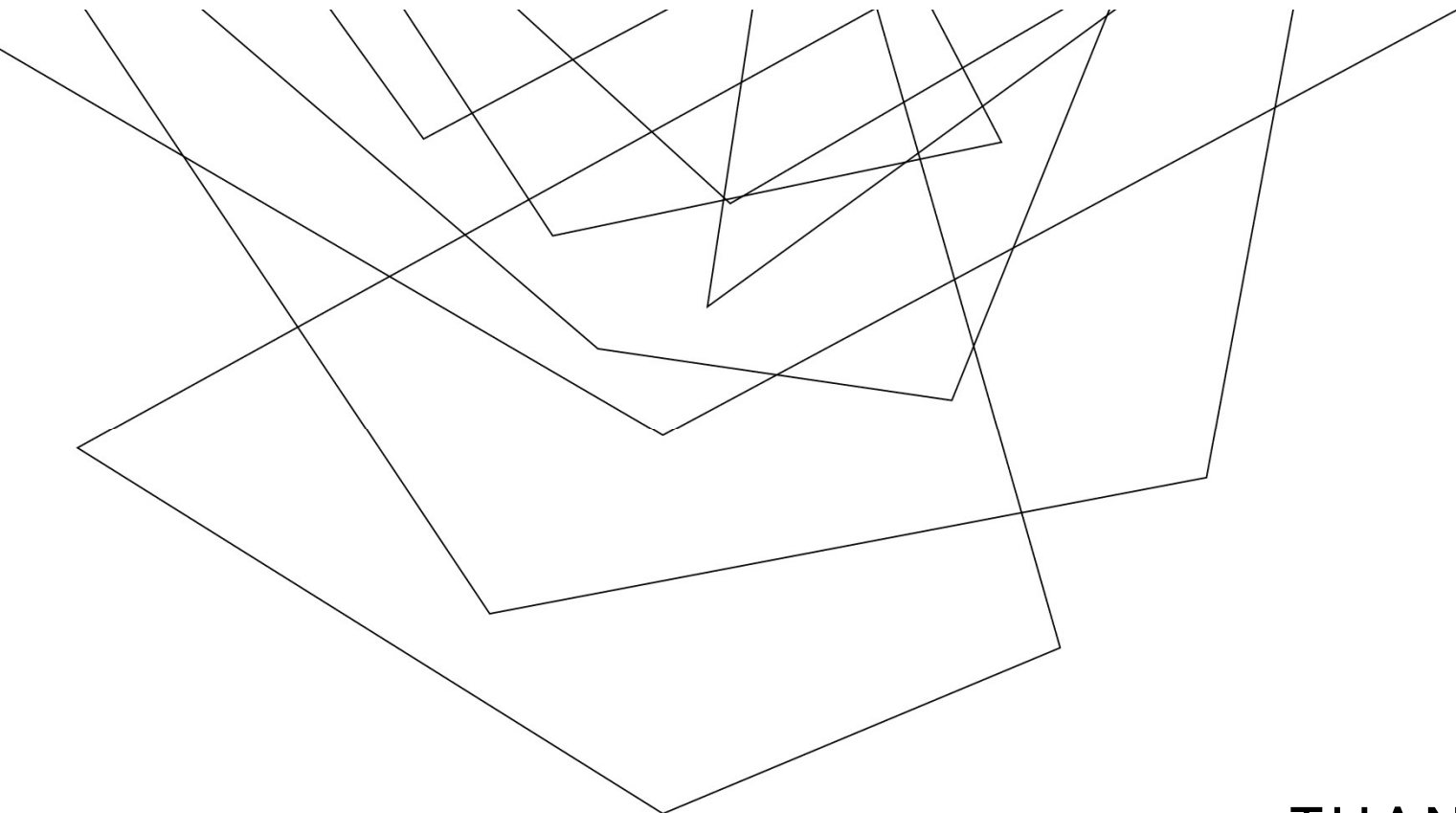
- Log Trading Volume = 0.095, not significant
- Log Market Cap = -0.121, not significant

Preferred Model: $R^2 = 0.124$

- Log Trading Volume = 0.295, not significant
- Log Market Cap = -0.525, weakly significant at 10%
- Volume to Market Cap Ratio = 1.665, not significant

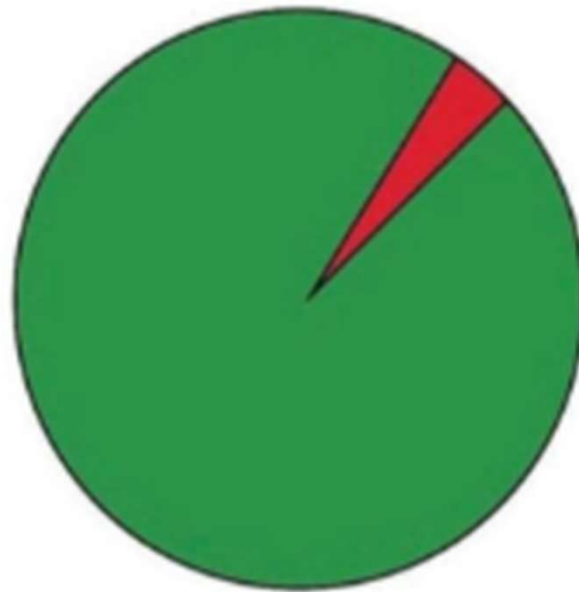
CONCLUSION

- The results give limited support that larger cryptocurrencies have lower volatility
- In the research model, higher market cost was significantly linked to lower volatility.
- In the filtered and preferred models, the variables were not statistically significant.
- Overall, cryptocurrency volatility is affected by more than just its market size.



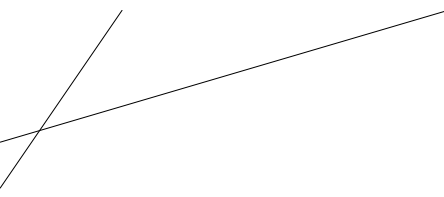
THANK YOU

Should I sell my Bitcoin?



■ NO
■ NO, BUT IN RED





SOURCES

1. Lecture 4
2. Lecture 6
3. Lecture 8
4. Lecture 9
5. Lecture 10
6. Lecture 11
7. <https://www.youtube.com/watch?v=JVQNYwo4AbU>
8. <https://www.youtube.com/watch?v=hpc5jyVpUpw>
9. <https://stackoverflow.com/questions/61977076/how-to-fetch-data-from-api-using-python>
10. <https://www.youtube.com/watch?v=NcffDnVBPHA>
11. <https://www.youtube.com/watch?v=ewXzgCow94g>
12. <https://htmlcolorcodes.com/>
13. <https://machinelearningplus.com/plots/python-scatter-plot/>
14. <https://www.youtube.com/watch?v=xcZib3qAc30>
15. <https://www.codecademy.com/learn/learn-linear-regression-in-r/modules/linear-regression-in-r/cheatsheet>